



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

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Branch: CSE (25-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No.: 7

TITLE: : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement:

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives:

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory:

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Outcomes:

Successfully implemented inheritance and interfaces in Java to achieve code reusability and abstraction. Applied these concepts to build real-world applications like a Shape calculator and an E-Commerce Product System.

```
interface Sound{
    void makeSound();
}

class Animal{
    String name;
    Animal (String n){
        this.name = n;
    }
    void eat(){
        System.out.println(name + " is barking.");
    }
}

class Dog extends Animal implements Sound{
    Dog(String n){
        super(n);
    }
    public void makeSound(){
        System.out.println(name + " barks: Woof!");
    }
}

public class JAVA_EXP_7{

    public static void main(String[] args) {
        Dog dog = new Dog("Puppy");
        dog.eat();
        dog.makeSound();
    }
}
```

```
orkspaceStorage\6b4136d8ec875b8bb324a9029103fc22\redhat.
eabel187\bin' 'JAVA_EXP_7'
Puppy is barking.
Puppy barks: Woof!
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```

Conclusion:

This assignment demonstrates how inheritance enables code reuse by allowing subclasses to acquire properties and methods from a parent class, while interfaces define contracts that ensure consistent behaviour across different classes. The Shape application and E-Commerce Product System successfully illustrate how these OOP concepts support abstraction, polymorphism, and modular software design in real-world scenarios.

Exercise:

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
class Shape {
    void area() {
        System.out.println("Area method not defined for generic shape.");
    }
}

class Circle extends Shape {
    double radius;
    Circle(double radius) {
        this.radius = radius;
    }
    void area() {
        double result = Math.PI * radius * radius;
        System.out.println("Circle Area: " + result);
    }
}

class Rectangle extends Shape {
    double length, width;
    Rectangle(double length, double width) {
        this.length = length;
        this.width = width;
    }
    void area() {
        double result = length * width;
        System.out.println("Rectangle Area: " + result);
    }
}

public class JAVA_EXP_7 {
    public static void main(String[] args) {
```

```

        Shape circle = new Circle(5);
        Shape rectangle = new Rectangle(4, 6);

        circle.area();
        rectangle.area();
    }
}

```

```

ppdata roaming (code (user workspace storage (00-123456789000
va\jdt_ws\JAVA_PRACTICALS_eabe1187\bin' 'JAVA_EXP_7'
Circle Area: 78.53981633974483
Rectangle Area: 24.0
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>

```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```

interface ProductInfo {
    void displayDetails();
}

class Product {
    String name;
    double price;
    Product(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

class Electronics extends Product implements ProductInfo {
    int warrantyYears;
    Electronics(String name, double price, int warrantyYears) {
        super(name, price);
        this.warrantyYears = warrantyYears;
    }
    public void displayDetails() {
        System.out.println("Electronics -> " + name + " | Price: " + price + " | Warranty: " + warrantyYears + " yrs");
    }
}

class Clothing extends Product implements ProductInfo {
    String size;
    Clothing(String name, double price, String size) {
        super(name, price);
        this.size = size;
    }
}

```

```

    }
    public void displayDetails() {
        System.out.println("Clothing -> " + name + " | Price: " + price + " | Size: " + size);
    }
}

class Grocery extends Product implements ProductInfo {
    String expiryDate;
    Grocery(String name, double price, String expiryDate) {
        super(name, price);
        this.expiryDate = expiryDate;
    }
    public void displayDetails() {
        System.out.println("Grocery -> " + name + " | Price: " + price + " | Expiry: " + expiryDate);
    }
}

public class JAVA_EXP_7 {
    public static void main(String[] args) {
        ProductInfo item1 = new Electronics("Laptop", 50000, 1);
        ProductInfo item2 = new Clothing("T-Shirt", 500, "M");
        ProductInfo item3 = new Grocery("Pulse Packet", 600, "Nov-2027");

        item1.displayDetails();
        item2.displayDetails();
        item3.displayDetails();
    }
}

```

```
eabel1187\bin' 'JAVA_EXP_7'
```

```
Electronics -> Laptop | Price: 50000.0 | Warranty: 1 yrs
```

```
Clothing -> T-Shirt | Price: 500.0 | Size: M
```

```
Grocery -> Pulse Packet | Price: 600.0 | Expiry: Nov-2027
```

```
PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>
```